

Ontology-Based Framework for Knowledge Representation, Semantic Reasoning, and Intelligent Information Access in Vrikshayurveda

Project Guide

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College Vision

Progress through education

College Mission

To seek, strive for and scale greater heights of quality education

Department Vision

To evolve as a center of excellence for higher education in the field of Artificial Intelligence and Machine Learning

Department Mission

M1: To equip graduates with the latest technologies in Artificial Intelligence and Machine Learning

M2: To imbibe new knowledge by collaborating with industry and research organizations to mould competent technocrats and entrepreneurs

M3: To instill societal and ethical responsibilities in all professional activities

- ① Motivation
- ② Objective
- ③ Literature Review
- ④ Methodology
- ⑤ Architecture / Block Diagram
- ⑥ Challenges Faced
- ⑦ Future Scope
- ⑧ References

- **Unstructured and Underutilized Traditional Knowledge** Vrikshayurveda knowledge primarily exists in textual form, making it difficult to computationally process, query, or analyze in a structured manner.
- **Lack of Semantic Modeling for Indian Agricultural Texts** There is limited implementation of ontology-driven semantic frameworks for representing plant–disease–treatment relationships in traditional agricultural literature.
- **Limitations of Keyword-Based Information Retrieval** Traditional search systems rely on keyword matching and fail to capture conceptual relationships and logical associations within the knowledge domain.

- **Need for Dynamic Semantic Query Access** A structured ontology requires a query-driven mechanism for context-aware knowledge retrieval.
- **Lack of Visual Knowledge Representation** Absence of graphical semantic visualization reduces interpretability of interconnected entities.
- **Digital Preservation of Indian Knowledge Systems** A semantic framework supports structured preservation and modernization of Vrikshayurveda knowledge.

OBJECTIVES

- To design and develop a formal ontology using Protégé and OWL 2 DL for structured representation of Vrikshayurveda knowledge.
- To model domain concepts including plants, diseases, soil, seasons, and treatments using semantic relationships and constraints.
- To validate ontology consistency and enable logical inference using the HermiT reasoner.
- To implement SPARQL-based query processing for dynamic and concept-driven knowledge retrieval.
- To develop an interactive knowledge graph visualization for explainable semantic representation.

Paper 1 (Base Paper): Ontology-Based Document Recommendation System Using Topic Modeling

IEEE 15th International Conference on Semantic Computing (ICSC), 2021

Objective:

- To improve document recommendation beyond keyword-based search.
- To integrate Topic Modeling (LDA/LSA) with ontology-based semantic reasoning.
- To enable semantic-level document similarity and intelligent retrieval.

Method:

- Dataset: Stack Exchange (Coffee domain).
- Data preprocessing (stopword removal, cleaning, stemming).
- Applied LDA and LSA for topic extraction.

- Optimized LDA using Grid Search (5 topics selected).
- Constructed ontology using Protégé (OWL).
- Used SPARQL querying (rdflib) for semantic retrieval.
- Developed front-end using Python (Tkinter).

Result:

- LDA produced more coherent and meaningful topics than LSA.
- Word Intrusion evaluation validated topic quality.
- System achieved improved semantic document recommendation.
- Demonstrated integration of Machine Learning + Ontology framework.

Paper 2: Ontology-Driven Knowledge-Based Health-Care System Challenges and Opportunities – Indian Scenario

Dr. Sunitha Abburu, Suresh Babu Golla
ISPRS Archives, Volume XL-8, 2014

Objective:

- To develop a centralized semantic knowledge base for public healthcare.
- To establish semantic relationships between diseases, places, and environmental factors.
- To support intelligent decision-making using ontology-driven inference.

Method:

- Collection of heterogeneous health data from hospitals.
- Standardization using EMR and EHR formats.
- RDF representation of health records.

- Development and merging of domain ontologies (Disease, Place, Environment).
- Ontology mapping and correlation using expert knowledge and statistical analysis.
- Implementation using Java, Apache Jena API, Oracle Semantic Store.
- SPARQL for semantic querying.

Result:

- Successfully built an integrated semantic knowledge base.
- Enabled spatial and non-spatial semantic queries.
- Demonstrated correlation between disease, environment, and location.
- Supported intelligent healthcare decision-making.

Paper 3: Ontology-Based Neuro-Symbolic AI Effects on Prediction Quality and Explainability

Smirnov et al., IEEE Access, 2024

Objective:

- To address the lack of interpretability in deep neural networks.
- To integrate domain ontologies into neural architectures using a neuro-symbolic approach.
- To analyze the impact of ontology integration on prediction accuracy and explainability.

Method:

- Proposed a computational framework combining:
 - Base Neural Network for classification.
 - Interpretation layer mapping hidden features to ontology concepts.
 - Logical block enforcing ontology axioms (subclass, equivalence, disjointness).

- Incorporated three loss components:
 - Target classification loss.
 - Concept supervision loss.
 - Logical consistency loss.
- Evaluated on XTRAINS and SCDB datasets using ROC-AUC.

Result:

- Self-explaining ontology-aware model achieved comparable or improved accuracy over standard classifiers.
- Ontology constraints enhanced concept-level prediction consistency.
- Semantic loss improved performance in semi-supervised settings.
- User study showed improved understandability and trust in model decisions.

Paper 4: An Ontology and Rule-Based Clinical Decision Support System for Personalized Nutrition Recommendations in NICU

Ravneet Kaur et al., IEEE Access, 2023

Objective:

- To develop an ontology-driven clinical decision support system (CDSS) for providing personalized nutrition recommendations to premature neonates in NICU.
- To ensure adherence to institutional neonatal nutrition guidelines using semantic reasoning.
- To automatically detect deviations from recommended feeding protocols.

Method:

- Developed a Nutrition Recommendation Ontology (NRO) using OWL.
- Constructed 121 classes, 366 axioms, and 157 semantic rules.
- Clinical data collected from 601 neonates and 8460 patient-days (2019–2021).
- Institutional nutrition guidelines converted into IF–THEN rule logic.

- Integrated ontology and rules into a reasoning engine.
- Used SPARQL queries for generating personalized feeding recommendations.
- Validated recommendations using expert neonatologists.

Result:

- The system achieved 98% accuracy when validated by neonatologists.
- Successfully identified guideline adherence and non-compliance.
- Enabled patient-specific feed recommendations based on:
 - Gestational age
 - Day of life
 - Risk factors
 - Clinical signs
 - Morbidities
- Demonstrated strong potential for improving nutrition delivery in NICU.

Paper 5: Plato – A Semantic Data Cube System Using Ontology-Based Data Access

IEEE Access, 2024

Objective:

- To enable semantic querying over large datasets through an Ontology-Based Data Access (OBDA) framework.

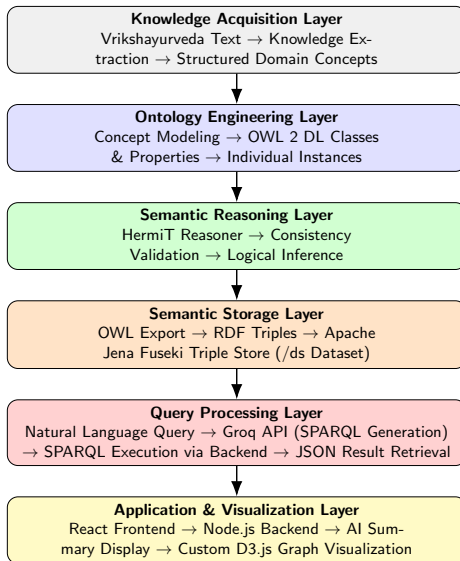
Method:

- Implementation of an OBDA layer connecting ontology with relational data
- Use of Ontop engine to translate SPARQL queries into SQL
- Concept-based query execution instead of direct database querying

COMPARITIVE ANALYSIS

Aspect	Paper1	Paper2	Paper3	Paper4	Paper5	Proposed Work
Problem Focus	Semantic document retrieval	Healthcare data integration	Explainable AI	Clinical decision support	Semantic data access	Modeling Vrikshayurveda knowledge
Ontology Usage	Post-processing layer	Integration framework	Neural constraint layer	Rule-based backbone	Data access abstraction	Core domain representation
Reasoning Type	SPARQL querying	RDF inference	Logical loss constraints	IF-THEN rules	SPARQL $\rightarrow$ SQL rewriting	OWL 2 DL reasoning (Hermit)
ML Dependency	LDA/LSA	No ML	Deep Neural Network	Rule-based	No ML	Logic-driven (no ML dependency)
Key Contribution	Improves similarity	Enables interoperability	Improves interpretability	Personalized decisions	Abstracts data complexity	Structured semantic modeling + logical retrieval

ARCHITECTURE / BLOCK DIAGRAM



1. Knowledge Acquisition & Structuring

- Extraction of domain knowledge from Vrikshayurveda textual corpus
- Identification of core entities (Plant, Disease, Soil, Season, Treatment)
- AI-assisted extraction of semantic relationships
- Classification and structuring of concepts for ontology modeling

2. Ontology Development (Protégé – OWL 2 DL)

- Construction of hierarchical class structure
- Definition of object and data properties
- Implementation of domain–range constraints
- Creation of individuals for validation

3. Semantic Reasoning

- Integration of HermiT reasoner
- Ontology consistency verification
- Inference of implicit semantic relationships

4. AI-Enabled SPARQL Query Processing

- Natural language query conversion using Groq API
- Execution of generated SPARQL over Apache Jena Fuseki
- Concept-driven semantic retrieval using ontology reasoning

5. Knowledge Graph Visualization & Interface

- Interactive visualization of ontology relationships
- Natural language query interface
- Graph-based semantic exploration and answer display

Ontology Implementation

- Developed using **Protégé**
- Modeled using OWL 2 DL in Protégé
- Semantic class hierarchy designed
- Domain–range constraints defined

Ontology Statistics

- Total Classes: **20**
- Total Object Properties: **15**
- Total Data Properties: **6**
- Total Individuals: **88**
- Reasoner Used: **HermiT**

IMPLEMENTATION OVERVIEW

Component	Tool / Technology
Ontology Design	Protégé + OWL 2 DL
Semantic Reasoning	HermiT Reasoner
Triple Store	Apache Jena Fuseki (/ds dataset)
NL to SPARQL Conversion	Groq API (LLM-based)
Backend Server	Node.js + Express
Frontend Interface	React (JavaScript/JSX)
Graph Visualization	Custom D3.js Graph (React-integrated)
Query Language	SPARQL 1.1

ONTOLOGY IMPLEMENTATION (PROTÉGÉ)

The screenshot displays the Protégé ontology editor interface for the 'vrushaayurveda' ontology. The top menu bar includes File, Edit, View, Reasoner, Tools, Refactor, Window, and Help. The address bar shows the ontology URL: <http://www.iish.org/ontology/vrushaayurveda/1.0>.

The left sidebar contains several panels:

- Class hierarchy: Tree**: A tree view showing the ontology structure. The 'Tree' class is selected under the 'Plant' class.
- Individuals**: A list of individuals, including 'Arjuna', 'Ashoka', 'Atimuktaka', 'Banana', 'Cane', 'Citron', 'Cluster Fig', and 'Cobra Saffron'.

The main workspace is divided into several panels:

- Annotations: Ashoka**: Shows annotations for the 'Ashoka' individual, including 'rdfs:label' (Ashoka) and 'rdfs:comment' (Beautiful and medicinal tree for gardens/houses).
- Description: Ashoka**: Shows the description of the 'Ashoka' individual, including 'Types' (Tree) and 'Same Individual As'.
- Property assertions: Ashoka**: Shows property assertions for the 'Ashoka' individual, including 'growsIn' (Mellow Fertile Soil), 'plantedInSeason' (Rainy Season), and 'suitableForLocation' (Park).

The bottom status bar indicates 'No Reasoner set. Select a reasoner from the Reasoner menu' and 'Show Inferences'.

SEMANTIC REASONING & VALIDATION

The screenshot displays the VUKSHAAYURVEDA ontology editor interface. The top menu bar includes File, Edit, View, Reasoner, Tools, Refactor, Window, and Help. The address bar shows the ontology URL: <http://www.ish.org/ontology/vrukshaayurveda/1.0>.

The interface is divided into several panes:

- Class hierarchy: Tree**: A tree view showing the ontology structure. The root is **owl:Thing**, which branches into **VrukshaayurvedaEntity** and **Plant**. **VrukshaayurvedaEntity** includes **Disease**, **Location**, **Material**, **ModernScientificNote**, **Nakshatra**, and **Plant**. **Plant** includes **Creepers**, **Shrub**, **Tree**, **PlantingPractice**, **Ritual**, **Season**, **Soil**, **Symptom**, and **TreatmentTechnique**.
- DL query**: A text area for entering a query. The query entered is `Plant and growsIn value 'Wet Lowland Soil'`. Below the text area are buttons for **Execute** and **Add to ontology**.
- Query results**: A list of instances returned by the query. The results are: Cluster Fig, Monkey Jack, Rose Apple, Wood Apple, Arjuna, Jackfruit, Kadamba, Pomegranate, Pongam, and Vanira.
- Query for**: A section with checkboxes for selecting query types: ☐ Direct superclasses, ☐ Superclasses, ☐ Equivalent classes, ☐ Direct subclasses, ☒ Subclasses, and ☒ Instances.
- Result filters**: A section with a text input for filtering results by name. It includes checkboxes for ☒ Display owl:Thing (in superclass results) and ☒ Disallow owl:Nothing.

The bottom status bar shows the Reasoner is active, the language is ENG IN, and the date is 02-03-2026.

SPARQL QUERY PROCESSING

The screenshot shows a web browser window with the address bar set to `localhost:3000`. The page title is "Vrikshayurveda". The navigation bar includes links for "Home", "About", "Graph", and "Assistant". The main content area displays the results of a SPARQL query.

Generated SPARQL:

```
PREFIX : <urn:absolute:h/absolute:h/>
SELECT ?practice ?name WHERE { ?practice a :PlantingPractice ; :auspiciousUnder :Pushya . OPTIONAL { ?practice :hasEnglishName ?name } }
```

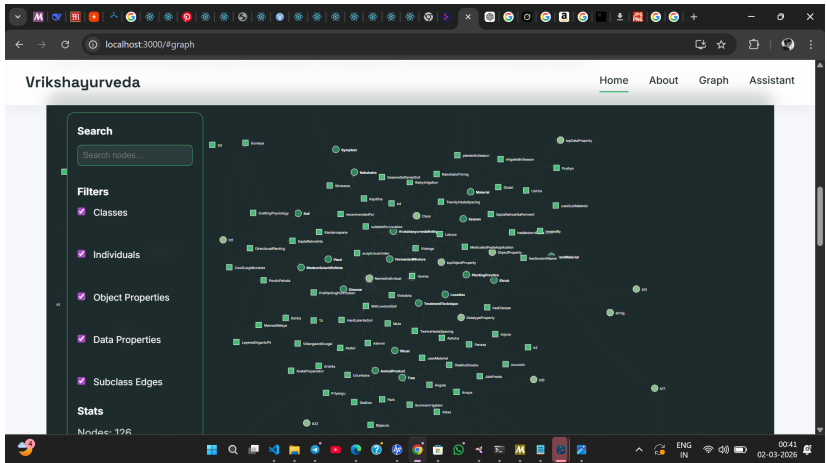
Raw Results:

```
{
  "head": {
    "vars": [
      "practice",
      "name"
    ]
  },
  "results": {
    "bindings": [
      {
        "practice": {
          "type": "uri",
          "value": "urn:absolute:h/absolute:h/TwentyMastaSpacing"
        },

```

The bottom of the image shows a Windows taskbar with various application icons and a system tray on the right displaying the date "02-03-2026" and time "00:41".

KNOWLEDGE GRAPH VISUALIZATION



RESULTS & SYSTEM OUTPUT

The screenshot shows a web browser window with the address bar displaying 'localhost:3000/#graph'. The page title is 'Vrikshayurveda'. The navigation bar includes links for 'Home', 'About', 'Graph', and 'Assistant'. The main content area is titled 'AI Assistant' and contains a chat interface. The user's question is: 'Planting practices associated with Pushya nakshatra are:'. The assistant's response is: 'Planting practices associated with Pushya nakshatra are NakshatraTiming, TwentyHastaSpacing, and PrePlantingRitual.' There is a 'Hide details' button below the answer.

Vrikshayurveda

Home About Graph Assistant

AI Assistant

Ask AI Assistant

Planting practices associated with Pushya nakshatra are:

Ask

Answer:

Planting practices associated with Pushya nakshatra are NakshatraTiming, TwentyHastaSpacing, and PrePlantingRitual.

Hide details

CHALLENGES FACED

1. Unstructured and Multilingual Source Content

- Vrikshayurveda text exists in narrative form without predefined structure
- Presence of Hindi/Sanskrit terminology alongside English
- Required selective extraction and semantic normalization

2. Knowledge Extraction and Concept Modeling

- Identifying domain entities and relationships from textual descriptions
- Avoiding ambiguity and redundancy in class hierarchy
- Structuring knowledge into ontology-compatible format

3. Ontology Design Complexity

- Designing balanced superclass–subclass hierarchy
- Defining precise domain–range constraints
- Maintaining logical consistency across entities

4. Connection & Endpoint Integration Issues

- Establishing stable communication between React frontend, Node.js backend, and Apache Jena Fuseki
- Handling CORS errors, incorrect dataset endpoints, and port mismatches
- Debugging SPARQL POST requests and JSON response parsing

5. AI-Assisted SPARQL Generation and Validation

- Translating natural language queries into syntactically correct SPARQL queries
- Preventing semantic mismatch between generated queries and ontology schema
- Validating and refining AI-generated queries before execution

1. Expansion of Knowledge Base

- Modeling the complete Vrikshayurveda corpus
- Inclusion of additional chapters and domain relationships
- Enrichment of ontology with extended agricultural concepts

2. Integration with Other Indian Knowledge Systems

- Extension to Ayurveda and traditional medicinal texts
- Development of a unified semantic framework for IKS domains

3. Semi-Automated Ontology Population

- NLP-based entity and relationship extraction
- Automated concept suggestion for ontology growth
- Reduction of manual modeling effort

4. Advanced Rule-Based Reasoning

- Integration of SWRL-based inference rules
- Implementation of conditional agricultural logic
- Enhanced semantic decision-support capabilities

5. Multilingual Ontology Support

- Inclusion of Hindi or Malayalam terminology
- Cross-language semantic mapping
- Bilingual semantic query capability

References

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Thank You